

Preamble

EART22001 Environmental Modelling

Environmental modelling is a core part of modern environmental science. Models allow us to combine physical, chemical and biological understanding with observations, explore processes that cannot easily be isolated in the real world, and test “what if?” scenarios. They are used in applications ranging from air-quality assessment and ecosystem dynamics to weather, climate and natural hazards.

Many environmental models are *dynamical models*: they describe how a system changes in time (and sometimes in space) using differential equations. A few simple equations have analytical solutions, but realistic environmental systems usually contain several interacting processes and must be solved approximately on a computer. The aim is not simply to obtain a numerical answer, but to understand what assumptions have been made, what the model can represent, and where its limitations lie.

Observations remain essential. They provide the evidence used to develop, constrain and evaluate models. However, environmental observations are often incomplete and we cannot control all variables in the way that we might in a laboratory experiment. Models therefore provide a complementary tool: they let us carry out controlled numerical experiments in which one process or parameter can be changed while others are held fixed.

Models are especially valuable when a real-world experiment would be impractical, very expensive, slow, or ethically problematic. One historical example is *marine cloud brightening*: the proposal to increase the reflectivity of low-level maritime clouds using sea-salt particles (Latham, 1990). Whether such an intervention would cool the climate as intended, have little net effect, or produce unintended consequences is exactly the kind of question that requires theory, observations and careful numerical modelling before any real-world intervention is considered. A model does not remove uncertainty, but it can make assumptions explicit and help us reason about consequences.

The first practical sessions therefore build a small computational toolkit. You will learn how to connect to a Linux server, work at the command line, obtain model code, edit input files, run calculations and inspect model output. You will use the same basic workflow repeatedly during the course, so the unfamiliar parts should become routine quite quickly.

What will I learn?

By the end of the module you should be better able to:

- explain why numerical models are used in environmental science and identify the assumptions behind a modelling approach;
- recognise the difference between ordinary differential equation (ODE), empirical, reduced-dimensional and fluid-dynamical models;
- use a remote Linux computing environment through SSH and transfer files securely;
- use basic Linux commands, Git/GitHub and, where useful, Docker;
- run existing environmental models, change model inputs and interpret the resulting output;
- use Python to analyse and visualise model data;
- connect mathematical ideas such as finite differences, forces, balances, transport and waves to environmental applications;
- evaluate the strengths, limitations and appropriate uses of different models.

Examples across the module include orbital dynamics, cloud and aerosol processes, pollutant transport, precipitation, and shallow-water dynamics. The emphasis is on using models to build physical understanding rather than treating them as black boxes.

What this course is not

This is not primarily a programming course and you are not expected to build a complete environmental model from scratch. We will look at equations, numerical ideas and selected pieces of code where they help explain how a model works, but the main goal is to understand modelling choices and to use existing models intelligently. Students who want to go further into numerical methods may find texts such as Hoffman (1992) useful.

References

Hoffman, J. D. (1992). *Numerical Methods for Engineers and Scientists*. McGraw-Hill.

Latham, J. (1990). Control of global warming? *Nature*, 347, 339–340. <https://doi.org/10.1038/347339b0>